

# Teaching Manual: Facts for 8 and 9

Scripted lesson | 60-75 minutes

## LESSON OVERVIEW

**Learning goal:** Use doubling and 10-minus-1 strategies for 8s and 9s.

### Success Criteria

- The student represents the idea with a model, drawing, table, or number line.
- The student uses the lesson vocabulary accurately.
- The student explains why a strategy works and checks reasonableness.
- The student can respond independently to: Show how  $4 \times 7$  can help solve  $8 \times 7$ .

### Teaching Theory

Harder facts should be derived from known facts through decomposition, doubling, and the distributive property. Division is most secure when linked to an unknown multiplication factor and interpreted in context.

**Instructional principle:** Move concrete -> pictorial -> abstract. Do not remove the model until the student can explain the symbolic work.

### Lesson Flow

|         |                               |
|---------|-------------------------------|
| 5-8 min | Number talk                   |
| 10 min  | Vocabulary and worked example |
| 20 min  | Main lesson                   |
| 15 min  | Practice                      |
| 10 min  | Challenge                     |
| 5 min   | Exit check                    |

# Materials: Preparation and Use

## BEFORE THE STUDENT ARRIVES

|                         |                                                                                                 |
|-------------------------|-------------------------------------------------------------------------------------------------|
| <b>Grid paper</b>       | Place beside the student. Use for drawings, organized records, and a second representation.     |
| <b>array cards</b>      | Cut and shuffle beforehand. Reveal only the cards needed for the current example.               |
| <b>mini whiteboard.</b> | Prepare before the lesson and introduce it only when it supports the current mathematical idea. |

## Table Setup

- Center: the main model or tool for today's lesson.
- Student side: packet, pencil, and only the materials currently needed.
- Teacher side: answer notes and observation record, kept out of student view.
- Leave a clear space where the student can point, move pieces, and explain.

## Vocabulary to Establish

**factor:** Ask the student for an example or picture before refining the definition.

**product:** Ask the student for an example or picture before refining the definition.

**quotient:** Ask the student for an example or picture before refining the definition.

**divisor:** Ask the student for an example or picture before refining the definition.

**remainder:** Ask the student for an example or picture before refining the definition.

**unknown factor:** Ask the student for an example or picture before refining the definition.

**Material rule:** The student should touch, draw, measure, or organize the mathematics before copying a procedure.

# Phase 1: Launch and Number Talk

## EXACT TEACHER LANGUAGE

|              |                                                                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Teacher says | "Today we are learning to use doubling and 10-minus-1 strategies for 8s and 9s. Before I show a method, I want to hear how you see the numbers." |
| Why          | Positions the student as a mathematical thinker and makes prior knowledge visible.                                                               |
| Listen for   | Useful representations, precise vocabulary, and whether the student checks reasonableness.                                                       |
| Then do      | Acknowledge the idea neutrally, then ask for evidence rather than immediately evaluating it.                                                     |

## Number Talk Prompt

|              |                                                                                     |
|--------------|-------------------------------------------------------------------------------------|
| Teacher says | "How can $10 \times 7$ help with $9 \times 7$ ?"                                    |
| Why          | Surfaces the relationships needed in the main lesson.                               |
| Listen for   | Any reasoning connected to multiplication and division, not only a final answer.    |
| Then do      | Ask: 'Can you show that another way?' Record two distinct strategies when possible. |

## Prompt Ladder

|                                                                                |
|--------------------------------------------------------------------------------|
| <b>First:</b> What do you notice? What could you try?                          |
| <b>Next:</b> Can you draw, build, or organize the information?                 |
| <b>Then:</b> Which part of today's vocabulary fits your idea?                  |
| <b>Last resort:</b> Solve a smaller related case, then return to the original. |

# Phase 2: Explicit Teaching

## MODEL, QUESTION, CONNECT

### Teaching Step 1

|                     |                                                                                 |
|---------------------|---------------------------------------------------------------------------------|
| <b>Teacher says</b> | "Let us find 8s by doubling a 4s fact. Tell me what I should do first and why." |
| <b>Why</b>          | Keeps the student cognitively active during explicit instruction.               |
| <b>Listen for</b>   | A connection between the representation and the mathematical symbols.           |
| <b>Then do</b>      | Model only the next move. Pause and return responsibility to the student.       |

### Teaching Step 2

|                     |                                                                                      |
|---------------------|--------------------------------------------------------------------------------------|
| <b>Teacher says</b> | "Let us find 9s as 10 groups minus 1 group. Tell me what I should do first and why." |
| <b>Why</b>          | Keeps the student cognitively active during explicit instruction.                    |
| <b>Listen for</b>   | A connection between the representation and the mathematical symbols.                |
| <b>Then do</b>      | Model only the next move. Pause and return responsibility to the student.            |

### Teaching Step 3

|                     |                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------|
| <b>Teacher says</b> | "Let us use arrays to justify both strategies. Tell me what I should do first and why." |
| <b>Why</b>          | Keeps the student cognitively active during explicit instruction.                       |
| <b>Listen for</b>   | A connection between the representation and the mathematical symbols.                   |
| <b>Then do</b>      | Model only the next move. Pause and return responsibility to the student.               |

### Teaching Step 4

|                     |                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Teacher says</b> | "Let us observe, but do not depend on, digit-sum patterns in 9s. Tell me what I should do first and why." |
| <b>Why</b>          | Keeps the student cognitively active during explicit instruction.                                         |
| <b>Listen for</b>   | A connection between the representation and the mathematical symbols.                                     |
| <b>Then do</b>      | Model only the next move. Pause and return responsibility to the student.                                 |

**Representation check:** Before leaving the model, ask the student to point to every part of the matching equation or statement.

# Phase 3: Guided and Independent Practice

## RELEASE RESPONSIBILITY GRADUALLY

### I Do -> We Do -> You Do

- I do: complete the packet's worked example while narrating the reason for each move.
- We do: solve the first practice task together; the student chooses the representation.
- You do: the student completes the remaining tasks while explaining selected answers.

|              |                                                                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Teacher says | "Practice task 1: $8 \times 6$ , $8 \times 7$ , $8 \times 9$ , $8 \times 12$ . Before solving, tell me what the task is asking and which tool might help." |
| Listen for   | Correct interpretation before computation.                                                                                                                 |
| Then do      | Use the prompt ladder. Avoid completing a step the student can perform.                                                                                    |

|              |                                                                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Teacher says | "Practice task 2: $9 \times 4$ , $9 \times 7$ , $9 \times 8$ , $9 \times 12$ . Before solving, tell me what the task is asking and which tool might help." |
| Listen for   | Correct interpretation before computation.                                                                                                                 |
| Then do      | Use the prompt ladder. Avoid completing a step the student can perform.                                                                                    |

|              |                                                                                                                             |
|--------------|-----------------------------------------------------------------------------------------------------------------------------|
| Teacher says | "Practice task 3: Write related division facts. Before solving, tell me what the task is asking and which tool might help." |
| Listen for   | Correct interpretation before computation.                                                                                  |
| Then do      | Use the prompt ladder. Avoid completing a step the student can perform.                                                     |

### Common Misconceptions

**Watch for:** Treating division only as sharing and missing grouping situations.

**Watch for:** Ignoring what a remainder means in the story.

**Watch for:** Using a fact trick without understanding why it works.

### Differentiation

**Secure:** Remove routine repetition and ask for a second method, proof, or generalization.

**Developing:** Keep the visual model visible and reduce the number size while preserving the same structure.

**Fragile:** Return to a concrete example, use one question at a time, and delay independent symbols.



# Phase 4: Challenge, Assessment, and Feedback

## PROTECT PRODUCTIVE STRUGGLE

|              |                                                                                                                                                                                                                       |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Teacher says | "Challenge: Find $9 \times 37$ mentally using the same strategy, then explain why it works. You may draw, make a table, test a simpler case, or work backward. I will give you thinking time before offering a hint." |
| Why          | Builds persistence and strategy selection rather than rapid answer production.                                                                                                                                        |
| Listen for   | An organized plan, revision after a failed attempt, and a justification.                                                                                                                                              |
| Then do      | Praise a specific mathematical action: organization, checking, representation, or explanation.                                                                                                                        |

## Exit Check

|              |                                                                                           |
|--------------|-------------------------------------------------------------------------------------------|
| Teacher says | "Complete this independently: Show how $4 \times 7$ can help solve $8 \times 7$ ."        |
| Why          | Provides evidence of the central lesson goal without teacher prompting.                   |
| Then do      | After completion, ask one clarifying question only if the written reasoning is ambiguous. |

## Evidence Guide

|            |                                           |                                                                        |
|------------|-------------------------------------------|------------------------------------------------------------------------|
| Secure     | Correct and independently justified       | Proceed; add a variation or ask the student to teach the idea.         |
| Developing | Correct with model or prompt              | Proceed with the model available and begin next lesson with retrieval. |
| Fragile    | Incorrect structure or unexplained answer | Reteach with smaller quantities and a concrete representation.         |



# Answer and Observation Guide

## USE AFTER INDEPENDENT WORK

### Expected Mathematical Evidence

- Number talk: a defensible response to 'How can  $10 \times 7$  help with  $9 \times 7$ ?' with a model or explanation.
- Main lesson: evidence that the student can use doubling and 10-minus-1 strategies for 8s and 9s.
- Practice: accurate work with units, labels, and a reasonableness check where appropriate.
- Challenge: an organized attempt at 'Find  $9 \times 37$  mentally using the same strategy, then explain why it works.' and an explanation of the chosen strategy.
- Exit: an independent response to 'Show how  $4 \times 7$  can help solve  $8 \times 7$ .'

### Teacher Verification Routine

- Rework every numerical answer before marking it.
- Accept a different method when the reasoning is valid.
- For open problems, verify that every condition is satisfied.
- For measurement, data, and geometry, require correct units and labels.
- For explanations, distinguish a correct statement from a demonstrated reason.

### Observation Record

#### A strong strategy the student used:

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#### A misconception to revisit:

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#### Vocabulary used precisely:

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#### Material or representation that helped:

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#### Plan for the next lesson:

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